CHE 396 Quiz 2

Alice Guzik

TOTAL POINTS

68 / 100

QUESTION 1

1 1 **20 / 20**

✓ **- 0 pts** Correct

- 20 pts wrong or not done

- 0 pts Click here to replace this description.

QUESTION 2

2 2 **3 / 20**

- 0 pts Correct

✓ **- 17 pts only LK and HK identified.**

- 20 pts wrong or not done

- 5 pts HK calculation is wrong.

QUESTION 3

3 3 **5 / 5**

✓ **- 0 pts** Correct

- 5 pts reflux ratio not mentioned.

- 2 pts reflux ratio stated as decreased.

QUESTION 4

4 4 **0 / 5**

- 0 pts Correct

✓ **- 5 pts** Incorrect

QUESTION 5

5 5 **5 / 5**

✓ **- 0 pts** Correct

- 5 pts Incorrect

QUESTION 6

6 6 **5 / 5**

✓ **- 0 pts** Correct

- 5 pts Incorrect

QUESTION 7

7 7 **5 / 5**

✓ **- 0 pts** Correct

- 5 pts incorrect

QUESTION 8

8 8 **0 / 5**

- 0 pts Correct

✓ **- 5 pts incorrect**

QUESTION 9

9 9 **0 / 5**

- 0 pts Correct

✓ **- 5 pts Incorrect**

QUESTION 10

10 10 **10 / 10**

✓ **- 0 pts** Correct

- 10 pts Incorrect

QUESTION 11

11 11 **15 / 15**

✓ **- 0 pts** Correct

- 15 pts Wrong or not done

- 10 pts Not explained well

- 5 pts Pressure not mentioned

- 5 pts C4s are not mentioned

CHE 396 SENIOR DESIGN Quiz 2 10/15/2019

Alice Guzik

Name

1. Determine the minimum utility requirements for the process set out below. Take the minimum approach temperature as 10°C. Design a heat exchanger network to achieve maximum energy recovery. Pinch point for the problem is 145°C.

	Actual temperature (ºC)		Heat capacity flow
Stream No.	Source	Target	rate *CP* (kW/ºC)
1	250	40 hot	1.5
2	140	230 cold	3
3	200	80 hot	2.5
4	20	180 cold	2

above pinch

above pinch

T_{pinch} H = 145 °C + $\frac{10}{2}$ → 150 °C

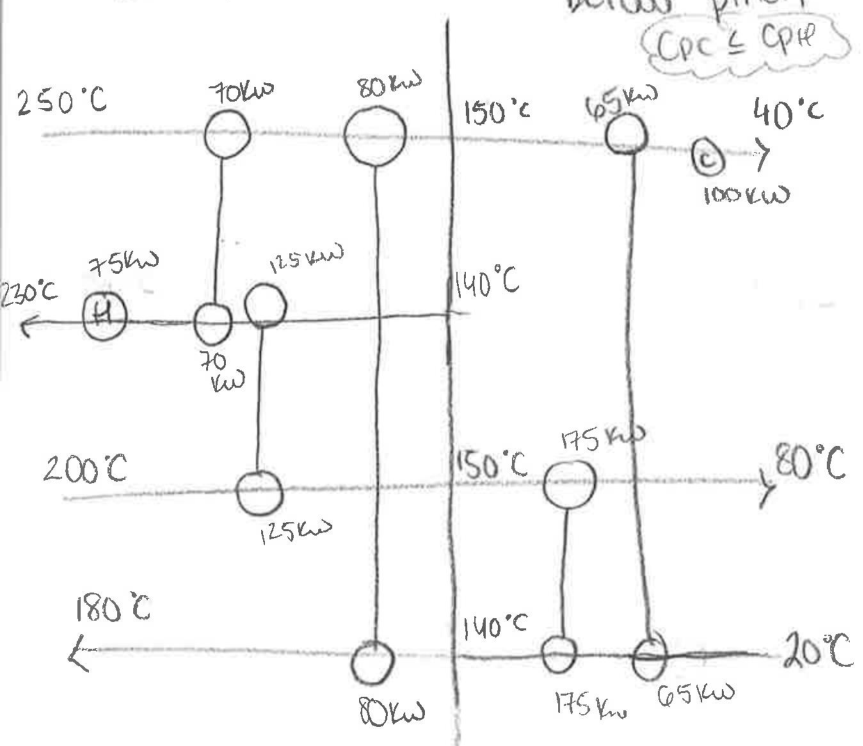
T_{pinch} C = 145 °C − $\frac{10}{2}$ → 140 °C

CpH ≤ CpC

Above pinch

below pinch

Cpc ≤ CpH



Above pinch ~~70 kw~~

stream① ΔH = 1.5 (250 − 150) = 150 kw

stream④ ΔH = 2 (180 − 140) = 80 kw

stream③ ΔH = 2.5 (200 − 150) = 125 kw

stream② ΔH = 3(230 − 140) = 270 kw

~~145 kw~~

75 kw

below pinch

Stream 3 ΔH = 2.5 (150 − 80) = 175 kw

stream 4 ΔH = 2.0 (140 − 20) = 240 kw

~~65 kw~~

Stream 1 ΔH = 1.5 (150 − 40) = 165 kw

$\boxed{100 \text{ kw}}$

$\left(\frac{175}{240}\right)2 = 1.46$ Cp

Split stream 4!

$\left(\frac{65}{240}\right) = 0.54$ Cp

2. The composition of the feed to a debutanizer is as follow; propane (C_3) is 10 kmol/h, isobutane (i-C_4) is 4 kmol/h, n-butane (n-C_4) is 6 kmol/h, isopentane (i-C_5) is 2 kmol/h, normal pentane (n-C_5)is 3 kmol/h and normal hexane (n-C_6) is 1 kmol/h. C_{5+} recovery in the bottoms is 99.5% and C_4 and lighter compounds recovery in the distillate is 99%. What is light key and heavy key recovery in the distillate?

light key → heaviest componet in top

heavy key → lightest componet in bottom

C_4 & lighter → $10 + 4 + 6 = \left(20 \frac{\text{kmol}}{\text{hr}}\right)$ → 99% top recovery

C_5+ → $2 + 3 + 1 = \left(5 \frac{\text{kmol}}{\text{hr}}\right)$ → 99.5% bottom recovery

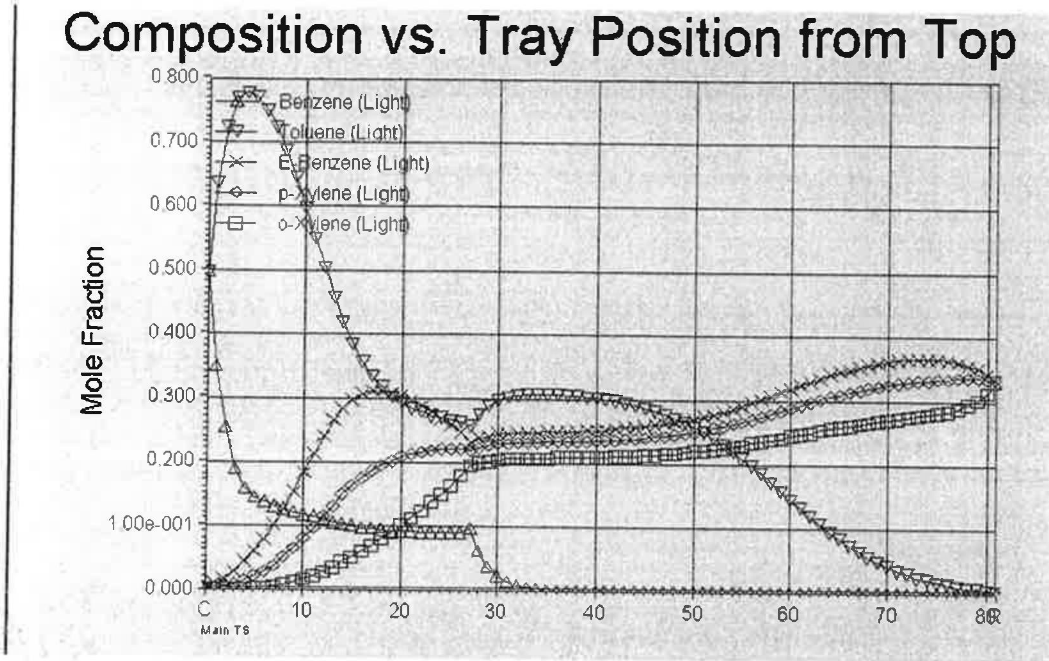
$19.8 \frac{\text{mol}}{\text{hr}}$ recovery ⇝ $\boxed{0.2 \text{ mol in bottoms}}$ ⇝ heavy key

$4.975 \frac{\text{mol}}{\text{hr}}$ recovery ⇝ $\boxed{0.025 \text{ mol in tops}}$ ⇝ light key

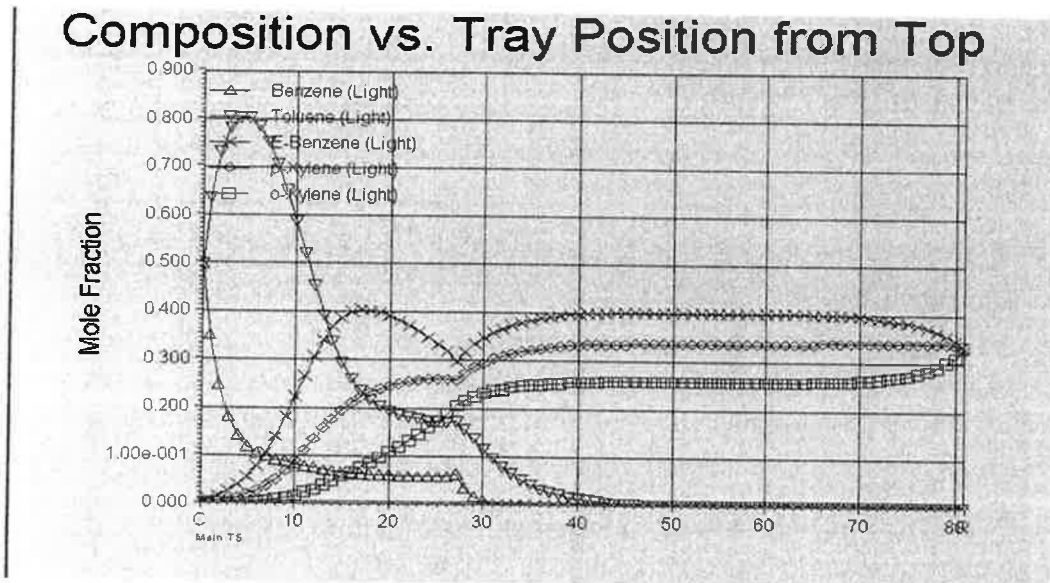
$\frac{0.2}{5} = 0.04 = \boxed{4\% \text{ heavy key in bottoms}}$

$\frac{0.025}{20} = 0.00125 = \boxed{0.125\% \text{ light key in tops}}$

3. Equimolar mixture of benzene, toluene, ethylbenzene (EB), orthoxylene (OX) and paraxylene (PX) is separated by distillation. Composition vs tray position was graphed as:



Later some parameters in the rigorous column has been changed and graph below is generated. Based on the graph below, which parameter has been changed, decreased or increased?



Feed tray → remained the same

# of trays → remained the same

either reflux ratio or operating pressure was increased

4. Distillation processes can only separate two species ("A" and "B") if:

A. They have similar molecular weights.
B. They have very different enthalpies of vaporization
C. They have different vapor pressures.
D. One species exists as a vapor (at STP), while the other species must be a liquid.

↳ standard pressure 1 ATM?

5.The pressure in a tray column

A. Is higher at the top.
B. Is higher at the bottom.
C. Depends on the type of tray and the tray design.
D. b and c.
E. None of the above.

6. The temperature in a tray column

A. Is higher at the top.
B. Is higher at the bottom.
C. Is the same on every tray.
D. Could be a or b, depending on the components involved.
E. None of the above.

7. If a column design results in one that is too tall, which of the following is a possible remedy?

A. Decrease the reflux ratio.
B. Reduce the number of trays.
C. Decrease the tray spacing.
D. b and c.
E. All of the above.

8.Which of the following is true about the feed to a distillation column?

A. The column feed determines the column pressure.
B. The column feed must be larger than the column pressure. → there is a pressure drop in the column
C. The column feed tray is always in the middle of the column.
D. b and c.
E. None of the above.

↳ should be near the middle to make sure there is equal stripping & refinate areas

9. Which of the following is true about the feed location in a distillation column?

A. Feed on the middle tray.
B. The feed should be at the tray with approximately the same temperature.
C. The feed should be at the tray with approximately the same composition.
D. b and c.
E. All of the above.

because you are trying to reach equilibrium at each stage

10. What was the pressure of MTBE column in your mini-project 2 design?

19 ATM ~> 19.24 bar

11. Methanol boiling point is higher than MTBE boiling point at 1atm. And they form azeotrope. However, in your mini-project problem 2, MTBE is separated from methanol by distillation and MTBE is the bottom product. Explain how that column works.

Changing the pressure of the column changes boiling point.

Also adding the extra C_4 componets can help in breaking the MTBE & Methanol azeotrope because the Methanol could form an azeotrope with the C_4s.

This also changes the boiling point